

Confidence for Which Prediction? Supervision Coverage Shapes Grounding Reliability

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Abstract

001 A grounding confidence score must distinguish correct
002 boxes from both mislocalizations and absent targets. Does
003 supervising that complete success event suffice? We study
004 matched confidence heads while fixing every output box
005 of two grounders. Correct-output labels improve correct-
006 versus-wrong ranking but need not reduce complete-output
007 risk; reading the selected query does not resolve this
008 tradeoff. A fixed-budget coverage intervention reveals a
009 more consequential distinction. Expanding positive super-
010 vision from L1 to all difficulty levels lowers correct-output-
011 head AUGRC by 0.332 points on FineCops, yet raises
012 existence-head risk by 0.916 points, reversing target prefer-
013 ence. Three-state decomposition explains the improvement
014 as better correct-box/no-target ordering bought at a cost
015 to correct/wrong-box ordering, not a uniform recovery of
016 correctness. The frozen correct-output head also improves
017 gRefCOCO risk by 0.154 points. These findings connect
018 confidence supervision to its training population and show
019 why better existence detection alone cannot establish more
020 reliable grounding outputs.

021 1. Introduction

022 A visual grounding system can return a plausible box for the
023 wrong object, or for an object that the expression does not
024 identify at all. These are different failures of the same out-
025 put decision. Generalized grounding makes both observable
026 by allowing expressions with no referent [3, 7, 8]. A useful
027 confidence score must retain correct localizations ahead of
028 both failure types.

029 The apparent solution is to supervise exactly this event:
030 the referent exists *and* the returned box is correct. An accu-
031 rate estimate of its conditional probability would optimally
032 order successes at fixed coverage under equal failure costs.
033 But a supervision label does not by itself give such an es-
034 timate. The head learns from a particular population of
035 requests, and the evidence available to its score can differ

from the evidence selecting the output box. We ask: *how* 036
do the supervision event and the positive training popula- 037
tion together shape the reliability of a fixed grounder? 038

We make this question experimentally separable from lo- 039
calization improvement. For each of two frozen grounders, 040
every confidence route returns exactly the same Native 041
box. Matched heads differ only in whether a present-but- 042
wrong request is labelled valid or unsuccessful. Under 043
a commonly used pairing construction, positive head su- 044
pervision reaches only L1 parents of edited negative ex- 045
pressions. On FineCops, correct-output supervision im- 046
proves correct/wrong-box ranking for both grounders, but 047
improves complete risk only for MDETR. A score can there- 048
fore become better at one relevant comparison without mak- 049
ing the overall output decision more reliable. 050

Two controlled tests sharpen this observation. First, read- 051
ing the query that actually supplies the output box does 052
not remove MM-GDINO’s disadvantage. Second, we vary 053
which positives receive supervision, keeping the head, neg- 054
ative requests, update budget and readout fixed. Expand- 055
ing from uniformly sampled L1 positives to the complete 056
positive set *reverses the preferred supervision target* on 057
FineCops. The Y head improves by 0.332 AUGRC points, 058
while the E head worsens by 0.916. Both directions occur in 059
all three seeds and persist in frozen transfer to gRefCOCO. 060
This is not merely a larger gap between two methods: the 061
absolute responses of both heads identify what produced the 062
reversal. 063

The explanation lies in the comparisons a complete out- 064
put score must make. Let C be a correct box, W a wrong 065
box for an existing referent, and N a no-target request. Ex- 066
panding supervision improves Y’s C/N ordering, especially 067
for positives outside the original difficulty coverage, but re- 068
duces its C/W ordering. The first benefit outweighs the sec- 069
ond cost at the observed mixture. For E, the C/W deterio- 070
ration is much larger: existence AUROC improves even as 071
complete risk worsens. Thus broader supervision is conse- 072
quential, but it does not simply repair every notion of cor- 073
rectness. 074

Our contribution is a controlled empirical finding about 075

What does confidence learn to trust?

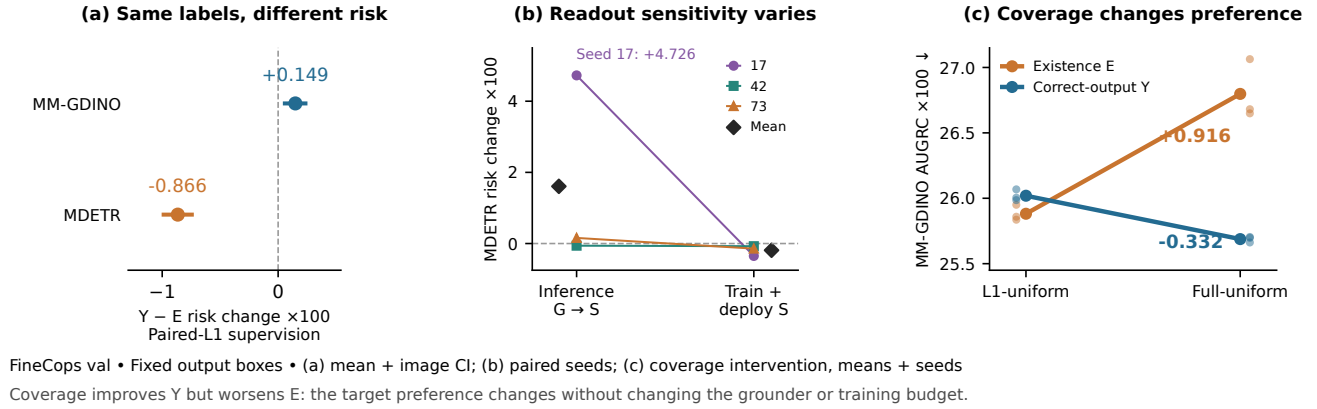


Figure 1. **The training population changes what confidence learns to trust.** All panels use FineCops validation and fixed Native boxes. (a) Under paired-L1 supervision, changing existence labels (E) to correct-output labels (Y) has opposite complete-risk effects on two grounders. (b) MDETR inference-only readout switching is seed-sensitive: seed 17 drives the large mean. Lines show paired seeds and diamonds their means, not image-bootstrap uncertainty. (c) A matched coverage intervention on MM-GDINO reduces Y risk but increases E risk; lines join means and small dots show seeds. Training budget and readout stay fixed. AUGRC is multiplied by 100; lower is better. Only (a) shows image-bootstrap intervals, conditional on the three trained heads.

grounding confidence: *the effect of the supervision event depends on the positive population used to learn it*. A matched coverage intervention establishes this dependence; readout controls test a natural alternative explanation; and three-state analysis connects the observed ranking changes to risk and its transfer. Fixed score combinations provide a practical closing comparison, and a record-only evaluator makes the analysis reusable. Together these results make the positive supervision population an explicit part of confidence design, rather than treating the label choice alone as a specification of what the score will reliably distinguish.

2. Related Work

Grounding and output quality. Grounding DINO [9] and MDETR [5] produce language-conditioned candidates, while GREC/gRefCOCO and FineCops extend evaluation beyond a guaranteed single valid referent [3, 7, 8]. Localization-aware detection scores [6] and learned reject options [1] connect confidence to prediction quality. Our study holds localization fixed to examine which examples teach a confidence head to distinguish a correct grounding output from each failure.

Joint failure detection. Invalid-input detection and mis-prediction detection are established as distinct problems [4]. OpenMix [14], SCOD [10], and SIRC [13] address aspects of joint failure detection, score combination and changing failure mixtures. AUGRC provides a generalized-risk view

of selective prediction [12]. These works already motivate evaluating both failures rather than only input validity. Our additional evidence concerns a concrete learning intervention: with grounding predictions and head training otherwise matched, changing labels and the positive supervision population changes which failure comparisons are learned, including after frozen transfer. The three-state identities explain these experiments; they are not presented as a new general framework.

Generalization and calibration. Calibration concerns the interpretation of scores as probabilities, including under distribution shift [2, 11]. Here we study ordering rather than fit an operating threshold. Grounding adds a useful distinction to the supervision population: an expression can refer to an existing object while the localizer returns the wrong box. The same image can support valid and no-target requests, so positive expression difficulty and image identity both enter our analysis. Frozen transfer tests the learned score without changing the source recipe.

3. Controlled Study

3.1. Fixed predictions, targets and readouts

For request (I, e) , a frozen localizer produces query features q_j , Native scores s_j^0 , and boxes b_j . Its unmodified selector chooses j^* and emits $b^* = b_{j^*}$. C denotes a present referent with Native IoU ≥ 0.5 , W a present referent with

an incorrect Native box, and N an absent referent. The existence target E is one on C and W; the correct-output target Y is one only on C. Negatives have no localization target.

Every head computes $z_j = h([q_j; s_j^0])$ from frozen inputs. We compare

$$c_G = \max_{j \in V} z_j, \quad c_S = z_{j^*}, \quad (1)$$

using the declared readout in both training and matched inference. Global and Selected heads start from the same initialization; we also switch readout at fixed weights as a separate diagnostic. Confidence never participates in box selection.

The two localizers are a FineCops positive-source MM-GDINO-T and the official RefCOCO MDETR-R101 EMA model. They supply 900 and 100 256-d queries, respectively, and each supplies its own Native-correctness labels. Architectures and adaptation histories differ, so effects are compared within localizer rather than attributed to architecture alone.

3.2. Two matched intervention blocks

Target $\times$ readout. Both localizers use the original 80,451 positive/edited-negative pairs. Their positive parents comprise 43,979 unique L1 requests: L2/L3 positives do not enter head loss. The full 83,341-positive training set is available, and was used for MM-GDINO’s earlier localizer adaptation, but not for these paired head updates. This distinction separates head coverage from representation exposure.

Target $\times$ positive coverage. On the same MM-GDINO, we fix Global-max and train E/Y with either all 54,015 L1 positives or all 83,341 positives, including 25,282 L2 and 4,044 L3. Both use uniform sampling over unique positives. The new L1-uniform control matters: the original paired recipe also omits 10,036 L1 examples and weights parents by their number of negative edits. Those changes are not attributed to broader difficulty coverage. Negatives keep their original order and frequency; only the positive stream changes between the two new populations.

All comparisons use three matched head seeds 17/42/73, a 50,179-parameter LayerNorm/MLP head, and 12,575 FP32 AdamW updates at LR 10^{-4} , zero weight decay and clip 0.1. Each update contains up to 32 positives and 32 negatives. Each source contributes half the mean BCE plus logit L2 of weight 10^{-3} ; Y receives no class rebalancing. Within a coverage condition, E/Y share initialization and sample order. Each new head sees 402,255 positive and 402,255 negative presentations, with every unique positive included. Expanding coverage changes the positive population and its weighting, not the total budget. Detailed schedules and all seed endpoints are in the supplement.

3.3. Evaluation and uncertainty

FineCops validation contains 9,426 positives and 9,029 text negatives. Frozen transfer uses gRefCOCO TestA/B: 11,563 single-target positives and 9,121 no-target requests on 1,500 images; 14,579 multi-target expressions are excluded. The FineCops train/val source-disjoint sensitivity retains 1,277 images, 9,848 positives and 7,716 negatives. This is not a claim of disjointness from all pretraining or MDETR’s RefCOCO adaptation. FineCops Test and negative images are not evaluated anew, and no evaluation data select a checkpoint or deployment threshold.

Mixed AUGRC integrates the probability of accepting a failed output over the fraction of requests retained. It is the primary complete-output risk, with AURC, pairwise AUROCs, Native P@1 and diagnostic FPR95 retained. All reported risk and AUROC values are multiplied by 100. Each block uses 5,000 paired image-cluster bootstrap draws shared by its arms and three seeds; gRef is stratified by TestA/B. Metrics are recomputed per seed, then equally averaged. Image intervals condition on those trained heads; sample SD separately describes seed variation. FPR95 recomputes positive q05 within each replicate. These are post-hoc-motivated mechanism studies with recipes fixed before the corresponding training, not virgin held-out confirmations.

4. Which Failure Does Supervision Separate?

Under paired-L1 supervision, Y improves C/W AUROC by 11.654 points on MM-GDINO and 13.889 on MDETR. Yet its AUGRC effect is $+0.149 [+0.058, +0.235]$ on the former and $-0.866 [-0.988, -0.745]$ on the latter. Both lose existence AUROC. Native P@1 remains 77.36% and 65.56% for every confidence route. The different risk responses therefore concern ordering the same outputs, not recovering better boxes.

Reading the output query is not sufficient. Selected readout leaves MM-GDINO’s Y–E disadvantage at $+0.159 [+0.077, +0.241]$. For MDETR it reduces Y risk by 0.189 points, but E benefits similarly; neither model’s FineCops target-by-readout interaction resolves a Y-specific advantage. Fixed-weight switching is different again: MDETR inference-only G→S changes risk by $+4.726, -0.061, +0.158$ across seeds, whereas matched retraining gives $-0.349, -0.075, -0.143$. Figure 1(b) exposes the seed 17-driven mean rather than treating it as a stable spatial explanation. Query choice, competition and training gradients all change with the readout design.

Two success/failure comparisons determine complete risk. Let U_{CW}, U_{CN}, U_{WN} denote pairwise AUROCs

Table 1. Target and readout under paired-L1 supervision on FineCops validation. Risk is mixed AUGRC $\times 100$ (mean, sample SD); target effects are Y–E with paired 95% intervals. Lower risk and higher pairwise AUROC are better. Each confidence route preserves its localizer’s Native boxes.

Localizer	Read	Risk E	Risk Y	Δ risk	ΔU_{CW}	ΔU_{CN}
MM-GDINO-T	G	26.23 (0.07)	26.38 (0.04)	+0.149 [+0.058, +0.235]	+11.654 [+10.721, +12.641]	-3.523 [-3.884, -3.166]
MM-GDINO-T	S	26.20 (0.07)	26.36 (0.04)	+0.159 [+0.077, +0.241]	+10.724 [+9.850, +11.606]	-3.359 [-3.700, -3.019]
MDETR-R101	G	29.71 (0.27)	28.84 (0.13)	-0.866 [-0.988, -0.745]	+13.889 [+13.044, +14.743]	+0.290 [-0.265, +0.857]
MDETR-R101	S	29.52 (0.28)	28.65 (0.02)	-0.864 [-0.993, -0.737]	+14.622 [+13.664, +15.565]	+0.017 [-0.577, +0.618]

with half credit for ties, and let $a = \Pr(C \mid E = 1)$. Existence AUROC includes

$$AUC_E = aU_{CN} + (1 - a)U_{WN}. \quad (2)$$

The W/N term orders two failures. In contrast, at no-target fraction π , the fixed-output AUGRC difference is

$$\Delta R = -(1 - \pi)a[(1 - \pi)(1 - a)\Delta U_{CW} + \pi\Delta U_{CN}]. \quad (3)$$

For MM-GDINO’s paired Y–E change, the C/W gain contributes -0.532 risk points, but the C/N loss contributes $+0.681$, giving $+0.149$ overall. For MDETR, the corresponding terms are -0.818 and -0.047 . This accounts for opposite risk effects despite the common decline in existence AUROC. It also explains why a W/N improvement alone is not evidence of better successful-output ordering.

The deficit is concentrated beyond head supervision. In MM-GDINO, the paired target change improves L1 C/W by 7.085 points and L1 C/N by 0.224. The full C/N effect is nevertheless -3.523 , comprising a $+0.169$ within-level contribution and -3.692 cross-level contribution. All negatives have L1 parents, so the latter compares L2/L3 correct outputs with these no-target requests. Difficulty conditioning locates the deficit but cannot establish its cause. The coverage intervention tests whether changing the supervised population changes these comparisons.

5. Coverage Changes the Supervision Tradeoff

The reversal survives a uniform-L1 control. Using every L1 positive uniformly reduces the risks of both original heads, yet Y remains worse than E by $+0.138$ $[+0.053, +0.223]$ AUGRC points. The target disadvantage therefore remains after changing parent weighting and including the omitted L1 examples. With full-positive supervision, the Y–E gap becomes -1.110 $[-1.279, -0.951]$, now favoring Y (Table 2). The negative target-by-coverage interaction is 1.248 $[-1.366, -1.138]$.

The absolute changes are essential to interpret this reversal. Y risk changes by -0.332 $[-0.436, -0.235]$, while E risk increases by $+0.916$ $[+0.755, +1.078]$. All three seeds agree on both directions. The 1.248-point interaction is therefore not a 1.248-point improvement of Y: it combines a modest Y improvement with a larger E deterioration. This distinction gives the coverage experiment a more informative outcome than a winner-only comparison.

Broader coverage redistributes, rather than uniformly improves, ordering. Y gains $+4.412$ $[+3.956, +4.885]$ points in C/N AUROC, but loses 11.411 $[10.582, 12.237]$ in C/W. Its complete-risk improvement is thus not a recovery of correct/wrong discrimination. Equation 3 shows that the C/N gain outweighs that cost at the observed mixture. The C/N gain has a $+5.182$ cross-level contribution and a -0.770 within-level contribution; even L1-conditional C/N declines by 1.019 points. This is a localized change in which requests are trusted, not a uniform benefit across difficulty strata.

For E, C/W AUROC falls from 67.08 to 33.69 while C/N improves by 3.153 points. W/N rises even more, by 27.359 points. Existence AUROC consequently improves by 8.633 points while complete risk *increases*. The existence label does not distinguish C from W; broader supervision produces a score that is better at its validity task but less useful for retaining correct boxes. The measured ranking changes establish this consequence without requiring an unobserved spatial or representation-level cause.

Full coverage does not eliminate every residual loss. Y still trails E in FineCops C/N by 2.327 $[1.730, 2.948]$ points under Full supervision. Its larger C/W advantage makes it preferable at the observed mixture, but the relative risk can reverse when no-target requests dominate. The internal crossover moves from $\pi = 0.434$ $[0.401, 0.467]$ under L1-uniform to 0.769 $[0.721, 0.819]$ under Full-uniform (Fig. 2). Coverage changes the range of mixtures favoring each target, rather than proving that the two learned abilities are

Table 2. Positive supervision coverage changes target preference with fixed MM-GDINO boxes, Global-max readout and training budget. The four risk cells show mean (sample SD), and differences show paired 95% intervals, all $\times 100$. $D_Y = R_{\text{Full},Y} - R_{\text{L1},Y}$, $D_E = R_{\text{Full},E} - R_{\text{L1},E}$, $I = D_Y - D_E$. gRef uses frozen heads; its two populations overlap.

Population	L1 E	L1 Y	Full E	Full Y	D_Y	D_E	I
FineCops val	25.88 (0.06)	26.02 (0.04)	26.80 (0.23)	25.69 (0.02)	-0.332 [-0.436, -0.235]	+0.916 [+0.755, +1.078]	-1.248 [-1.366, -1.138]
gRef source-disjoint	28.65 (0.04)	28.49 (0.04)	29.69 (0.12)	28.33 (0.07)	-0.154 [-0.223, -0.083]	+1.043 [+0.917, +1.169]	-1.197 [-1.306, -1.088]
gRef Full	28.75 (0.04)	28.62 (0.05)	29.78 (0.12)	28.46 (0.07)	-0.157 [-0.222, -0.091]	+1.026 [+0.907, +1.141]	-1.183 [-1.281, -1.081]

Table 3. Which comparisons change when supervision expands from L1 to Full? Full–L1 effects $\times 100$ [paired 95% CI]. C/W and C/N determine complete risk; W/N compares two failures but contributes to existence AUROC. The target-specific absolute changes separate improvement in Y from degradation in E.

Population	Target	ΔU_{CW}	ΔU_{CN}	ΔU_{WN}	ΔAUC_E	Δrisk
FineCops val	Y	-11.411 [-12.237, -10.582]	+4.412 [+3.956, +4.885]	+18.281 [+17.300, +19.286]	+7.552 [+7.039, +8.076]	-0.332 [-0.436, -0.235]
FineCops val	E	-33.392 [-35.041, -31.809]	+3.153 [+2.494, +3.841]	+27.359 [+25.965, +28.812]	+8.633 [+7.895, +9.408]	+0.916 [+0.755, +1.078]
gRef source-disjoint	Y	-0.657 [-1.201, -0.118]	+1.483 [+1.202, +1.757]	+2.115 [+1.766, +2.472]	+1.761 [+1.518, +2.015]	-0.154 [-0.223, -0.083]
gRef source-disjoint	E	-10.814 [-11.955, -9.698]	-1.491 [-1.947, -1.045]	+3.189 [+2.659, +3.743]	+0.566 [+0.187, +0.959]	+1.043 [+0.917, +1.169]

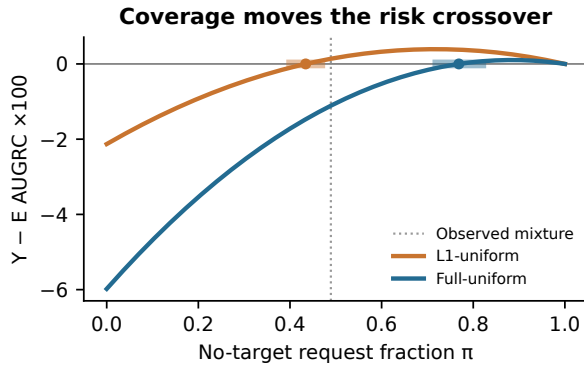


Figure 2. **The tradeoff moves; it does not vanish.** FineCops Y–E AUGRC under fixed class-conditional requests and varying no-target fraction. Curves use the estimated mean pairwise AUROCs, not a fitted deployment rule. Horizontal marks at zero show bootstrap intervals for the crossover location, not confidence bands for the curves.

now jointly optimized for every population. This identity applies to AUGRC; AURC is evaluated separately.

6. Transfer and Practical Consequences

The coverage response persists with frozen heads. On gRef source-disjoint requests, Full supervision changes Y risk by -0.154 [-0.223, -0.083] and E risk by +1.043

[+0.917, +1.169]; Full TestA/B gives -0.157 [-0.222, -0.091] and +1.026 [+0.907, +1.141], respectively. All seeds agree on these risk directions. For Y, C/N improves by 1.483 points while C/W decreases by 0.657 on the source-disjoint surface, repeating the direction of the source-side tradeoff with a smaller correctness cost. Both success comparisons deteriorate for E there, even though its mean existence AUROC increases slightly.

Full-coverage Y–E C/N is unresolved on gRef source-disjoint (+0.193 [-0.373, +0.777]), whereas Y’s C/W advantage is clear. We therefore report improved complete risk without claiming pairwise dominance. The two gRef surfaces overlap; they provide a source-exclusion sensitivity, not two independent replications. Coverage training was performed only on MM-GDINO. MDETR broadens the target/readout comparison, not the coverage intervention’s architecture scope.

Fixed score combinations offer setting-specific gains.

An alternative practical response is to combine already learned scores. On the original paired-L1 heads, we evaluate Product, $\log \max(s_0, 10^{-6}) + \log \sigma(c_{G,E})$, and a pre-specified SIRC-style rule [13]. Neither changes Native boxes; the latter uses only training-positive normalization statistics. Product improves MM-GDINO gRef source-disjoint AUGRC over Native, G/E and G/Y, including -0.130 [-0.187, -0.074] relative to G/Y. On FineCops it is worse than both learned heads. MDETR similarly shows reference- and population-dependent gains. Full tables are

in the supplement. These controls demonstrate that fixed scores can sometimes be combined profitably, not a new fusion method or dominance over the newly trained full-coverage heads.

7. Discussion and Conclusion

The central result is not that correct-output supervision is inherently conflicted, or that one readout repairs confidence. Its effect depends on the positive population used to learn it. With a fixed grounder and fixed budget, expanding that population improves the Y head but worsens the E head, changing the preferred target. The three-state decomposition explains this response through a concrete exchange between C/W and C/N ordering, and frozen transfer preserves the main risk directions.

For confidence design, this suggests an actionable sequence: specify the event to be trusted, make the positive supervision population explicit, and examine both success/failure comparisons before concluding that a validity gain improves grounding reliability. Our intervention changes positive examples, images and relative difficulty weights together; it is not an isolated causal effect of a difficulty label. The evidence covers two fixed localizers, with coverage intervention on one, three head seeds, and text-negative/single-no-target requests. Within that scope, it advances from observing different scores to identifying a training-population change that alters their error priorities and downstream risk.

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